

PROJECTS & PUBLICATIONS

4MATIV Transportation Reliability | Python, R, Pandas, Excel, Hypothesis Testing, IQR Anomaly Detection [Deck](#)

- Engineered a **weighted 5-component confidence scoring model** across **10,800+ trips** and **20+ vendors** to validate GPS-based route execution; uncovered a **5.8x deviation spread** in route adherence and flagged **47%** of low-confidence trips via **IQR anomaly detection**, directly influencing vendor contract renegotiations

CSI Bid Analysis, Carlson Analytics Lab | SAS, Tableau, Excel

- Synthesized **1,000+ bid records** in **Tableau** to surface pricing gaps, regional hotspots, and seasonal trends; uncovered a **25% YoY cost increase** and translated findings into impact scenarios scored against **business value criteria**, shaping roadmap commitments presented to the Field Advisory Board

InsideInsight: Airbnb Pricing Intelligence | Python, Spark, Databricks, LangGraph, LLaMA, Streamlit [GitHub](#)

- Architected a **Bronze/Silver/Gold medallion pipeline** on **Spark** and **Databricks** processing **1.95M rows**; deployed a **LangGraph multi-agent AI** with **LLaMA 3.3 70B** to surface pricing gaps, identifying listings underpriced by **65%** representing **\$51,830 in recoverable revenue** — awarded **Judges & Instructor's Favorite** at Carlson

Santander Customer Churn Prediction | Python, Scikit-learn, XGBoost, LightGBM, R [Deck](#)

- Developed a **6-model ensemble** on **76,020 customer profiles** with a **24:1 class imbalance** achieving **AUC-ROC 0.827**; quantified business impact as **\$256M in protected annual revenue** and presented a data-driven retention roadmap to senior management in a **financial services** context

AGI Hackathon: Green ROI Travel Emissions Optimization | Python, Tableau, Sustainability Analytics [Deck](#)

- Investigated **87K+ travel records** to quantify carbon footprint, identifying flights as **84% of total CO₂ emissions**; modeled a **Green ROI framework** demonstrating **4.5M+ kg CO₂ reduction** and proposed an **AI-driven nudge system** to guide sustainable corporate travel decisions

ExportEase: Global Reach, Seamless Collaboration | AWS, Python, HuggingFace, Django, Docker [Demo](#)

- Architected **Doc Buddy**, an **ML-powered** document analytics tool using **HuggingFace LayoutLM** and **AWS Translate** to automate real-time Q&A across **10+ document types**; reduced manual processing effort by **30%** and response time by **45%**, accelerating operations **4x** across **5+ countries**

Barclays ConnectCare | Python, NLP, AWS, GCP, QlikSense [GitHub](#)

- Designed **ML-driven NLP models** with **GCP Dialogflow** to classify and route customer support tickets, cutting critical issue resolution time by **45%**; built **AWS** data pipelines and **QlikSense** dashboards to monitor SLA trends, reducing latency by **35%** for **1.5K+ users**

Medicare Spending Drivers: Variation in Intensity and Frequency of Care | SAS, R, SQL, Tableau [Paper](#)

- Investigated **274,654 Medicare beneficiaries** using **SAS** with weighted statistical methods; established that **3.1%** of patients drive **46.7%** of total spending through episode intensity, delivering cost-containment recommendations to healthcare stakeholders via **Tableau** dashboards and a written white paper

ARMD-NetX: Ensemble Learning Network for ARMD Detection | OpenCV, TensorFlow [Paper](#)

- Developed a multimodal deep learning framework combining **CNNs** and **ensemble** meta-classifiers achieving **99.41% accuracy** on OCT images; published an [OCT-Fundus dataset](#) with **12K+ labeled images** to advance **explainable AI** in ophthalmology

Publications & Patent

- [IEEE CINS 2024](#) — **Lung Cancer Classification Using Deep Learning**; peer-reviewed publication applying deep learning models to multi-class lung cancer classification from medical imaging data
- [IEEE ICC 2025](#) — **Enhancing Bone Fracture Detection Using Hybrid VGG16-CNN Architecture**; proposed a hybrid transfer learning approach improving fracture detection accuracy on X-ray datasets
- **Patent No. 202421056179 — DoppelScan: Dual Authentication Contactless Attendance System**; filed patent for a computer vision-based dual-factor authentication system enabling contactless attendance tracking